0312

PAC

P: distribution of data

given $S = \{(\ x_i, c(x_i)\ )\}_{i=1}^m \sim P^n$

- $R_S(h) = \frac{1}{m}\sum_{i=1}^m 1_{h(x_i)\neq c(x_i)}$ (the experiment error)
- $R(h) = \mathbb{E}_{(x,c(x))\sim P}[1_{h(x)\neq c(x)}] = \mathbb{P}\{h(x) \neq c(x)\}$
  (for the binary classification)

if S are i.i.d. samples, $\mathbb{E}_S\{R_S(h)\} = R(h)$ (from the law of large number theorem)

Hoeffding's inequality - sum of independent bounded random variables

$S_m = z_1 + ... + z_m, a_i \leq z_i \leq b_i$

$\mathbb{P}(|S_m - \mathbb{E}[S_m]| \geq t) \leq 2e^{-\frac{2t^2}{\sum(b_i-a_i)^2}}$

Lemma 1.

$\mathbb{P}[|R_S(h) - \mathbb{E}[R_S(h)]| \geq \epsilon] \leq 2e^{-2m\epsilon^2}$

proof.

- $R_S(h) = \frac{1}{m}\sum_{i=1}^m 1_{h(x_i)\neq c(x_i)}$
  $= z_1 + ... + z_m$, where $z_i = \frac{1}{m}1_{h(x_i)\neq c(x_i)} \in [0, \frac{1}{m}]$
  $\leq 2e^{-\frac{2\epsilon^2}{m\times(\frac{1}{m})^2}}$

Let $h_1, ..., h_{\|H\|}$ be members in H

- $\mathbb{P}[|R_S(h) - R(h)| \leq \epsilon] \geq 1 - \delta$ is learned with $m \geq m(\epsilon, \delta)$

$\mathbb{P}[\exists h \in H : \underbrace{|R_S(h) - \mathbb{E}[R_S(h)]|}_{\text{generalization error}} \geq \epsilon] = \mathbb{P} \cup_{i=1}^{|H|} \{|R_S(h_i) - \mathbb{E}[R_S(h_i)]| \geq \epsilon\}$
$\leq \sum_{i=1}^{|H|} \mathbb{P}\{|R_S(h_i) - \mathbb{E}[R_S(h_i)]| \geq \epsilon\} \leq 2|H|e^{-2m\epsilon^2}$

Let $2|H|e^{-2m\epsilon^2} \leq \delta$, we need
$-2m\epsilon^2 \leq \log \frac{\delta}{2|H|} \implies m \geq \frac{1}{2\epsilon^2}(\log \frac{1}{\delta} \log(2|H|))$

Therefore, error is bounded if $m \geq m_0(\epsilon, \delta) = \frac{1}{2\epsilon^2}(\log \frac{1}{\delta} + \log(2|H|))$

for the consistent case, $h(x) = c(x)$, $m_0(\epsilon, \delta) = \frac{1}{2\epsilon}(\log \frac{1}{\delta} + \log(2|H|))$, the required number of data is only ϵ

$|R_S(h) - R(h)| \leq \sqrt{\frac{\log|H| + \log\frac{2}{\delta}}{2m}}$

$$\implies R(h) \leq \underbrace{R_S(h)}_{bias} + \underbrace{\sqrt{\frac{\log|H| + \log\frac{2}{\delta}}{2m}}}_{variance}$$

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PAC on decision tree

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assume there're n features (dimension of inputs) and m training points, then the number of leaf in a tree cannot greater than m.

Therefore, if H is the hypothesis class, then $|H| \leq m$ is finite. Besides, there's some inconsistent cases. These points give us a nice condition to use PAC analysis.

Assume $|H_i|$(the number of decision tree with i leaves)

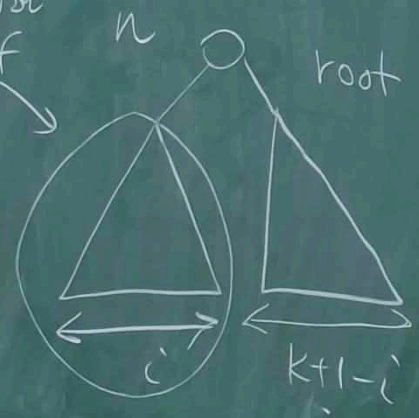
assume $|H_i|$ with $i \leq k$ is given, then how can we analyze $|H_{k+1}|$

we can imagine the following case

Let $|H_i|$ (the number of decision tree with i leaves)

Assume $|H_i|$ with $i \le k$ is given.

$|H_{k+1}|$ decision tree of i leaves

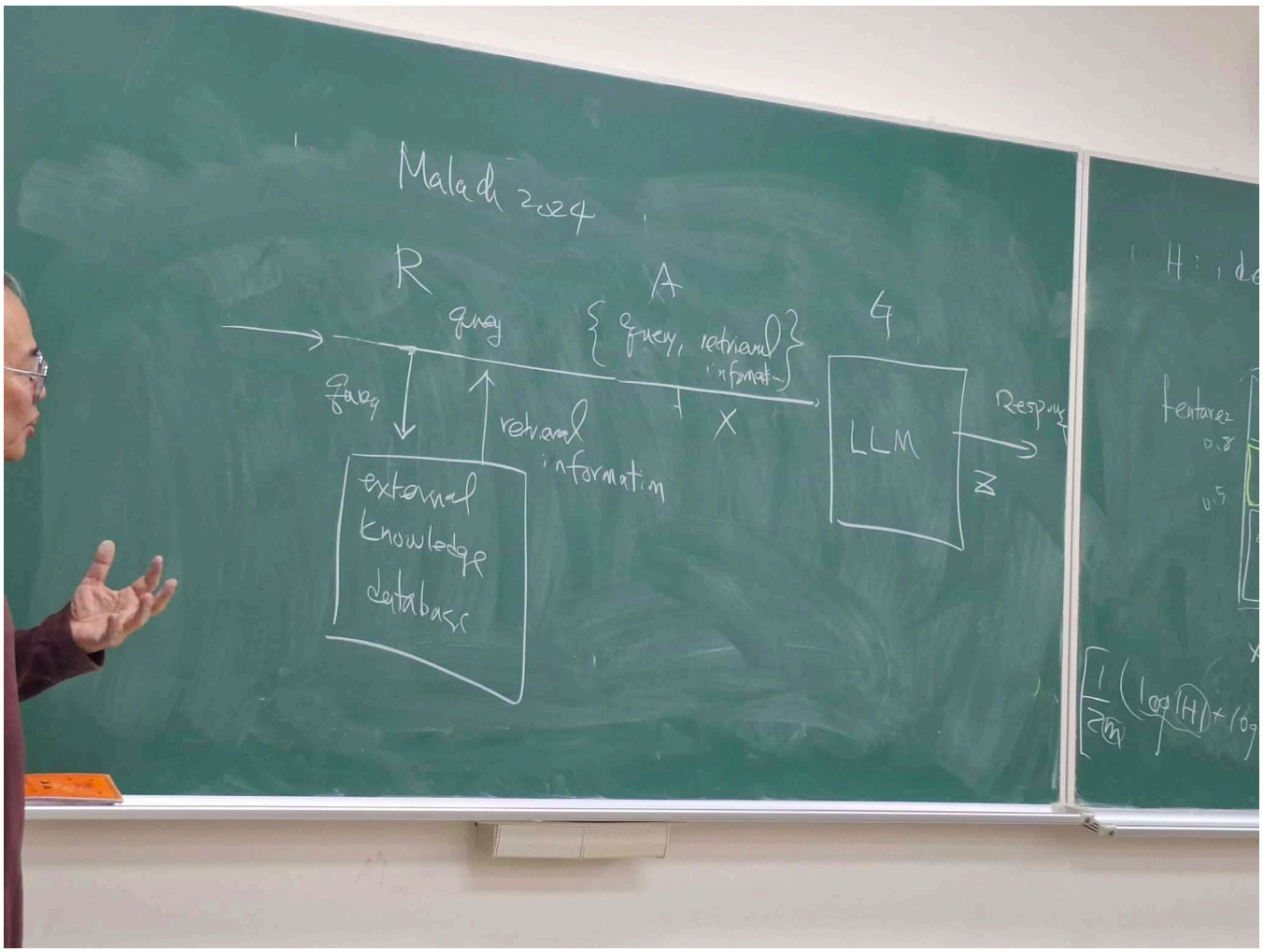


H_i

$|H_{k+1}| = n \sum_{i=0}^{k} |H_i| \times |H_{k+1-i}|$

then we can solved
$|H_k| = n^{k-1}(k+1)^{2k-1}$

analyze of RAG system



Formulation:

- D : finite set of token
- $X = D^n$: the space of strings of tokens

For some t, $Z_t = D^t$

Autoregressive function (AR) function h:

$$\underbrace{X}_{\text{prefix}} \times \underbrace{Z}_{\text{curerent generated tokens}} \rightarrow D \text{ (next-token)}$$

$H = \{h\}$: all autoregressive functions

Fixed T, generating T tokens.
For some distribution P over $X \times Z_T$, P is reliable by H if there exists $h \in H$ s.t. over $(x, z) \sim P, h(X, Z_{<t}) = Z_t \forall t \leq T$

- $Z = Z_1 Z_2 ... Z_T$
- $Z_{<t} = Z_1 Z_2 ... Z_{t-1}$
- h is a deterministic, accurateky products the next token for all prefix of $(X, Z) \sim P$

H is (PAC) AR-learnable
P is realizable by H, for every $(\epsilon, \delta) \in (0, 1)^2$, if $\exists m_0(\epsilon, \delta)$ s.t. h is trained by $m \geq m_0(\epsilon, \delta)$ i.i.d. samples such that with probability $\geq 1 - \delta$, we have $\mathbb{P}[\exists t \leq T | h(x, z_{<t}) \neq Z_t] \leq \epsilon$ for $(x, z) \sim P$

the last step: how to determine h

generate T supervised classifier:

- $h_1 : (x, \phi) \rightarrow z_1$, P_1 introduced from P, $m_0(\frac{\epsilon}{T}, \frac{\delta}{T})$
- $h_2 : (x, z_1) \rightarrow z_2$, P_2 introduced from P_1, $m_0(\frac{\epsilon}{T}, \frac{\delta}{T})$
- $h_T : (x, z_1 ... z_{T-1}) \rightarrow z_T$, P_T introduced from P_{T-1}, $m_0(\frac{\epsilon}{T}, \frac{\delta}{T})$

choose $h(x, z_{<t}) = h_t(x, z_{<t}) \forall t \leq T$

Fixed Tm generating T tokens.
if $H_1, ... H_T$ are PAC-learnable with sample complexity $m_0(\epsilon/T, \delta/T)$, then

$H = H_1 \times H_2 \times ... \times H_T$ is AR-learnable w.r.t sample complexity $m_0(\epsilon, \delta)$

proof:

$$P[(h(x, z_{<i}) \neq z_i) \leq \epsilon] \geq 1 - \delta/T$$

$$\mathbb{P}[\exists t \leq T | h(x, z_{<t}) \neq z_t] = \mathbb{P}[\cup_{t=1}^{T} (h_t(x z_t) \neq z_t)]$$
$$\leq \sum_i \mathbb{P}[h_i(x, z_{<i}) \neq z_i] \leq T \times \frac{\epsilon}{T} = \epsilon$$